CS & IT ENGINEERING

Operating System

CPU Scheduling

Lecture - 06

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Recap of Previous Lecture

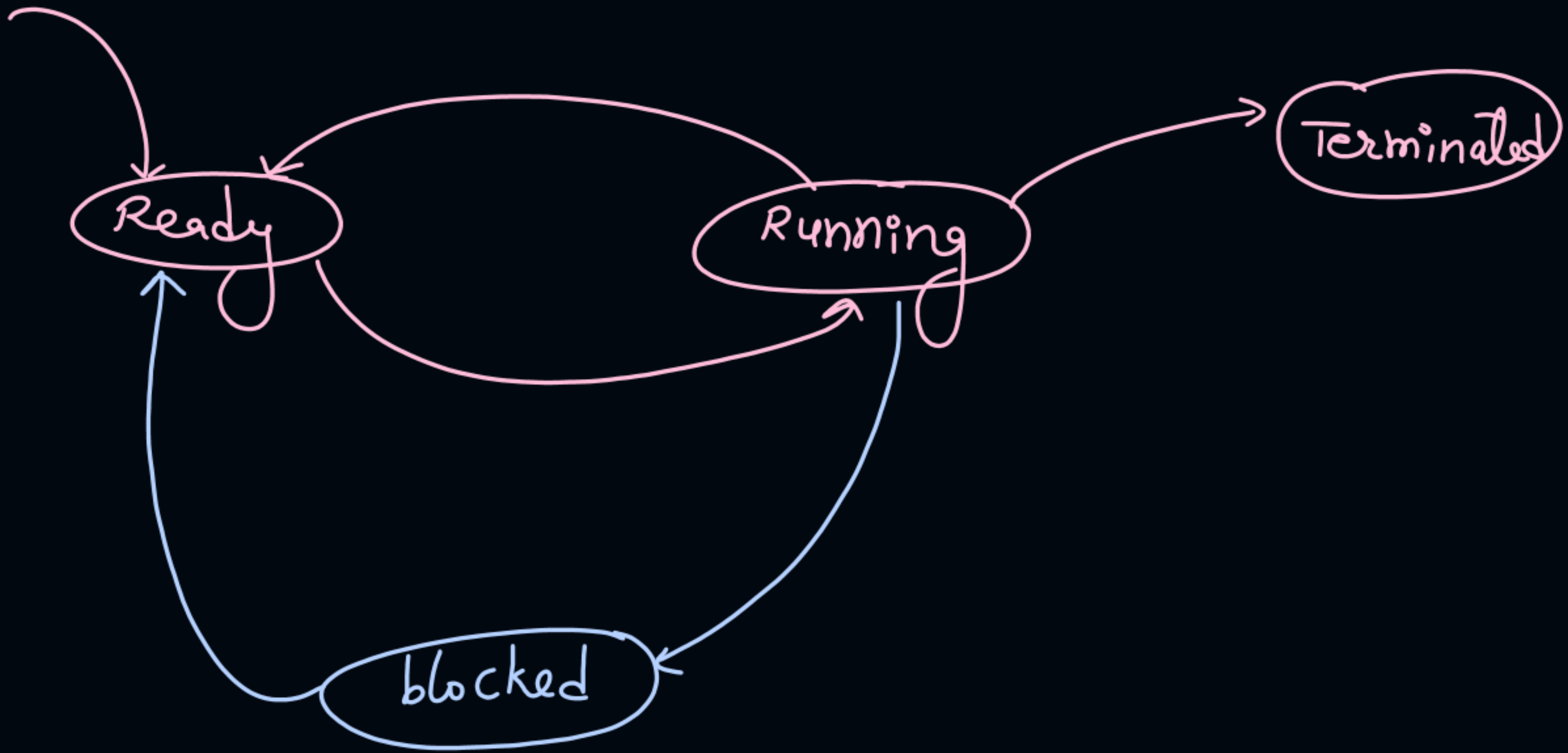
Topic Multilevel Queue Scheduling

Topic Multilevel Feedback Queue Scheduling

Topic Questions on Scheduling

Topics to be Covered

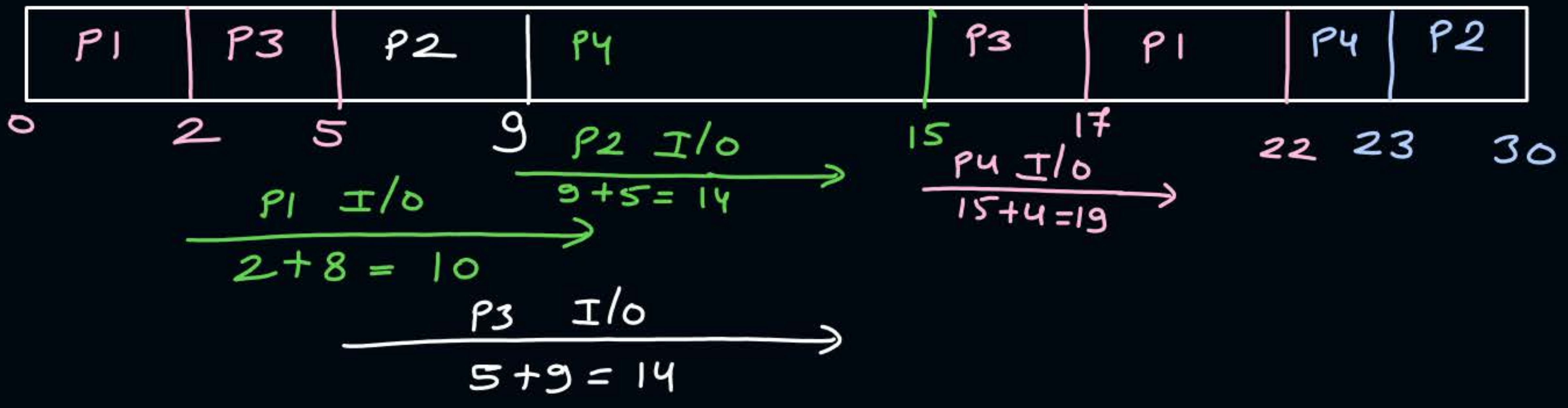
Topic Questions on Scheduling


Ready
Running
Terminated
blocked

[MCQ]

#Q. Consider a process scenario in which each process executes first in CPU then goes for IO operation, then once again process needs a CPU bursts and then terminates. Following is given a process scenario in which for CPU execution system uses non preemptive SJF algorithm. Consider system has enough number of resources to carry out IO operations for ~~only 2~~ all processes in parallel at a time. What is the average waiting time for the execution for the processes?

WT in ready queue	Process	Arrival Time	CPU Burst Time	IO Burst Time	CPU Burst Time	CT	TAT
22-(2+8+5)=7	P1	0	2	8	5	22	22
14	P2	0	4	5	7	30	30
3	P3	0	3	9	2	17	17
12	P4	0	6	4	1	23	23



Time	Ready Queue
0	P1(2), P2(4), P3(3), P4(6)
2	P2(4), P3(3), P4(6)
5	P2(4), P4(6)
9	P4(6)
15	P1(5), P2(7), P3(2)
17	P1(5), P2(7)

Time	Ready Queue
22	P2(7), P4(1)
23	P2(7)

Ques) SJF

	AT	CPU BT	I/O	CPU BT
P1	0	6	2	4
P2	0	2	4	3
P3	0	5	3	1

P2	P3	P2	P3	P1		P1
0–2	2–7	7–10	10–11	11–17	17–19	19–23

0 2 7 10 11 17 19 23

$\xrightarrow[2+4=6]{P2\ I/O}$

$\xrightarrow[7+3=10]{P3\ I/O}$

$\xrightarrow[17+2=19]{P1\ I/O}$

Time	Ready Queue
0	P1(6), P2(2), P3(5)
2	P1(6), P3(5)
7	P1(6), P2(3)
10	P1(6), P3(1)
11	P1(6)
17	No process
19	P1(4)

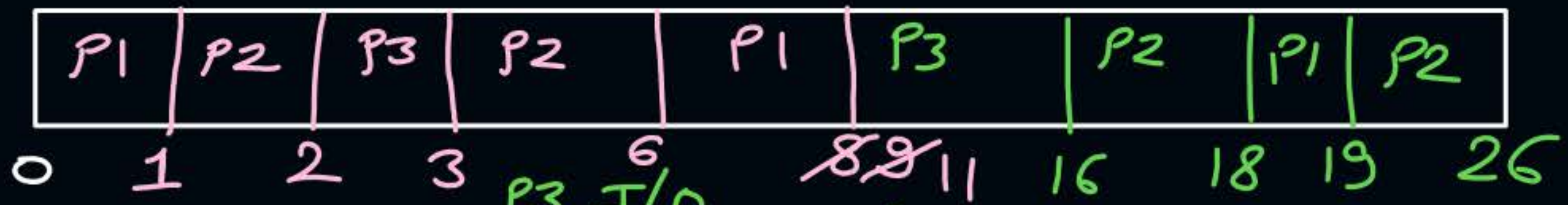
% of time CPU is idle $= \frac{2}{23} * 100\%$

$= 8.7\%$

[MCQ]

#Q. Consider a process scenario in which each process executes first in CPU then goes for IO operation, then once again process needs a CPU bursts and then terminates. Following is given a process scenario in which for CPU execution system uses preemptive SRTF algorithm. Consider system has enough number of resources to carry out IO operations for all processes in parallel at a time. What is the average waiting time for the execution for the processes?

Process	Arrival Time	CPU Burst Time	IO Burst Time	CPU Burst Time
P1	0	6	7	1
P2	1	4	2	9
P3	2	~~1~~	~~6~~	5



P3 I/O
$3+6=9$

P2 I/O
$6+2=8$

P1 I/O
$11+7=18$

P1	~~6~~ ~~5~~ ~~3~~ ~~2~~ ~~0~~
P2	~~4~~ ~~0~~ ~~9~~ 7
P3	0 ~~0~~

Time	Ready Queue
0	P1(6)
1	P1(5), P2(4)
2	P1(5), P2(3), P3(1)
3	P1(5), P2(3)
6	P1(5)
8	P1(3), P2(9)
9	~~P1(2)~~, P2(9), P3(5)
11	P2(9), P3(5)
16	P2(9)
18	P2(7), P1(1)
19	P2(7)

[MCQ]

#Q. The arrival time, priority and duration of the CPU and I/O bursts for each of three processes P1, P2 and P3 are given in the table below. Each process has a CPU burst followed by an I.O burst followed by another CPU burst. Assume that each process has its own I/O resource.

Process	AT	Priority	BT (CPU)	BT(I/O)	BT(CPU)
P1	0	2	1	5	3
P2	2	3(lowest)	3	3	1
P3	3	1(highest)	2	3	1

The multi - programmed operating system uses preemptive priority scheduling. What are the finish times of the process P1, P2 and P3 ?

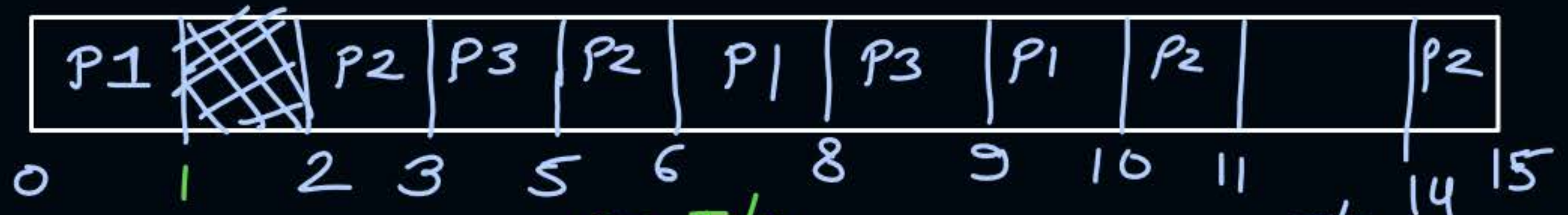
GATE - 2006

A 11, 15, 9

B 10, 15, 9

C 11, 16, 10

D 12, 17, 11



P3 I/O

$5+3=8$

P2 I/O

$11+3$
$=14$

P1 I/O

$1+5=6$

At time	Ready Queue
0	P1
1	—
2	P2
3	P2, P3
5	P2
6	P2, P1
8	P2, P1, P3
9	P1, P2
10	P2
11	—
14	P2

[MCQ]

H.W.

#Q. Multilevel Queue Scheduling, with fixed priority preemptive algorithm
Queue 1: RR with Q=2 (Higher priority)
Queue 2: SJF

Process	Arrival Time	Burst Time	Queue
P1	0	3	1
P2	1	3	2
P3	2	5	2
P4	1	4	1
P5	11	4	2
P6	15	3	1
P7	16	2	1

Topic : CPU Utilization

ex :-

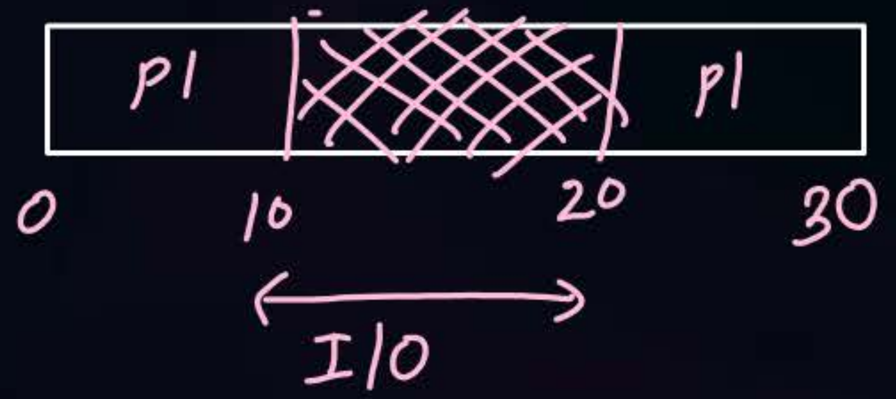
	BT
P1	10

1. Without IO Operations

$$\text{CPU utilization} = 1 \quad \text{or} \quad 100\,\%$$

2. With IO Operations

BT	I/O	BT
10	10	10



if p is fraction of time required for I/O of a process

if one process $\Rightarrow$ CPU utilization $= 1 - p$

if 2 —〃— $\Rightarrow$ —〃— $= 1 - p^2$

if 3 —〃— $\Rightarrow$ CPU utilization $= 1 - p^3$

if n —〃— $\Rightarrow$ —〃— $= \boxed{1 - p^n}$

[NAT]

#Q. A computer system has degree of multiprogramming 10. And maximum IO wait, that can be tolerated is 0.5. The CPU utilization is ___ %?

$$= 1 - (0.5)^{10}$$

$$= 0.99 \text{ or } 99\%$$

[NAT]

H.W.

#Q. Consider the following four processes with arrival times (in milliseconds) and their length of CPU bursts (in milliseconds) as shown below:

Process	P1	P2	P3	P4
Arrival time	0	1	3	4
CPU burst Time	3	1	3	Z

These processes are run on a single processor using preemptive Shortest Remaining Time First scheduling algorithm. If the average waiting time of the processes is 1 millisecond, then the value of Z is ________.

Gate - 2019

[MCQ]

#Q. A uni-processor computer system only has two processes, both of which alternate 10 ms CPU bursts with 90 ms I/O bursts. Both the process were created at nearly the same time. The I/O of both processor can proceed in parallel. Which of the following scheduling strategies will result in the least CPU utilization (over a long period of time) for this system ?

GATE - 2003

A First come first served scheduling

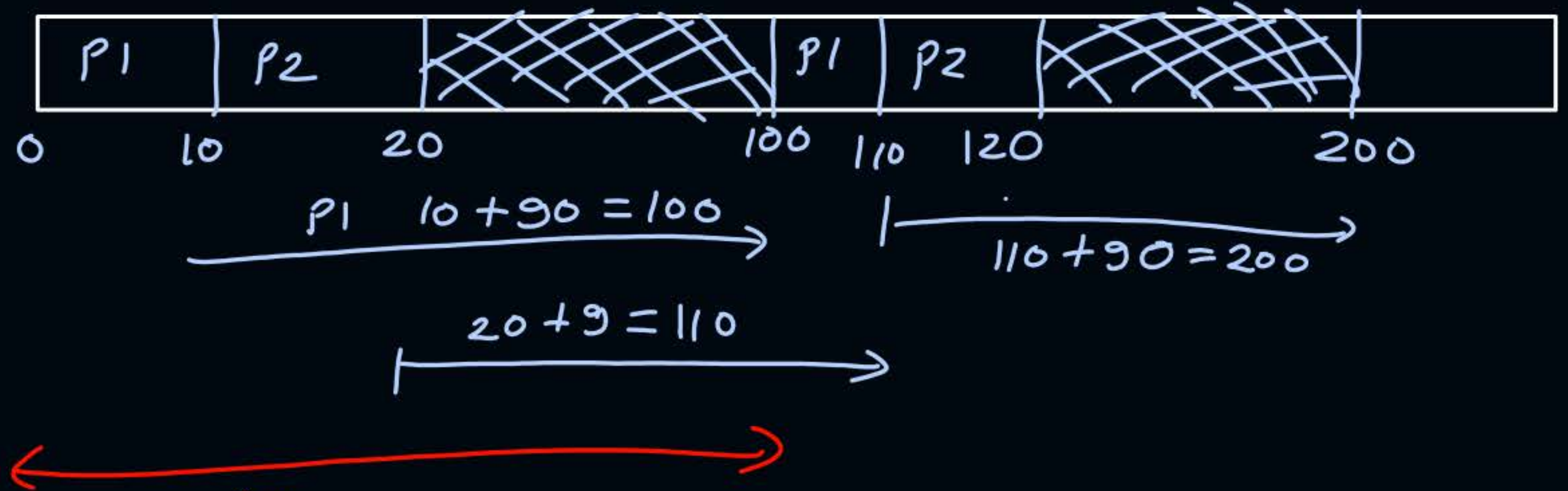
B Shortest remaining time first scheduling

C Static priority scheduling with different priorities for the two processes

D Round robin scheduling with a time quantum of 5 ms.

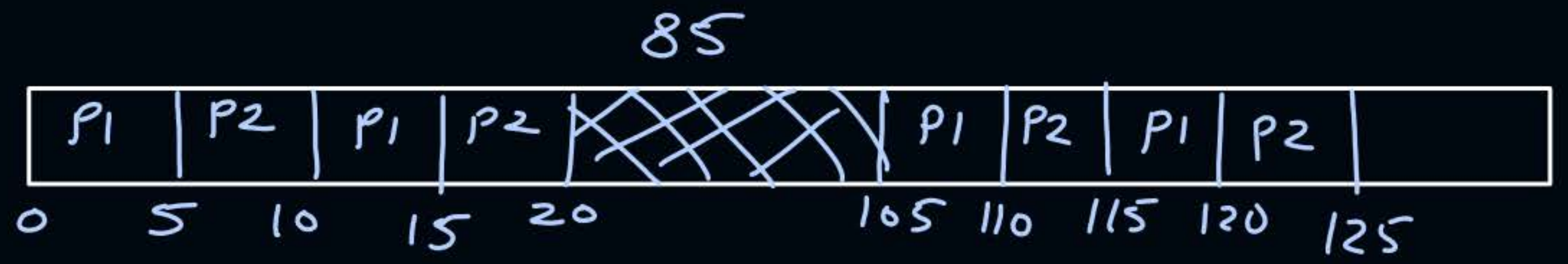
	CPU	I/O
P1	10	90
P2	10	90

FCFS, Pri, SRTF



CPU not utilized

$\frac{80}{100}$



$15 + 90 = 105$

$20 + 90 = 110$

$120 + 90 = 210$

$\frac{85}{105}$ CPU not utilized

[MCQ]

H. W.

#Q. Consider three processes, all arriving at time zero, with total execution time of 10, 20 and 30 units, respectively. Each process spends the first 20% of execution time doing I/O, the next 70% of time doing computation, and the last 10% of time doing I/O again. The operating system uses a shortest remaining compute time first scheduling algorithm and schedules a new process either when the running process gets blocked on I/O or when the running process finishes its compute burst. Assume that all I/O operations can be overlapped as much as possible. For what percentage of time does the CPU remain idle?

GATE - 2006

A 0%

B 10.6%

C 30.0%

D 89.4%

[NAT]

Ans = 12

#Q. Consider a uniprocessor system executing three tasks T_1,T_2 and T_3 each of which is composed of an infinite sequence of jobs (or instances) which arrive periodically at intervals of 3, 7 and 20 milliseconds, respectively. The priority of each task is the inverse of its period, and the available tasks are scheduled in order of priority, which is the highest priority task scheduled first. Each instance of T_1, T_2 and T_3 requires an execution time of 1, 2 and 4 milliseconds, respectively. Given that all tasks initially arrive at the beginning of the 1st millisecond and task preemptions are allowed, the first instance of T_3 completes its execution at the end of milliseconds.

GATE - 2015

Task	AT	BT
T1	0	1 ~~0~~
T2	0	2 ~~0~~
T3	0	~~4~~ ~~2~~
T1	3	1 ~~0~~
T1	6	~~1~~
T2	7	~~2~~
T1	9	~~1~~
T1	12	1
T2	14	2

T1	T2	T1	T3	T1	T2	T1	T3	

0 1 3 4 6 7 9 10 12

Time	Ready Queue
0	T1, T2, T3
1	T2, T3
3	T3, T1
4	T3
6	T1, T3
7	T3, T2

	R.Q.
9	T3, T1
10	T3

2 mins Summary

Topic Questions on Scheduling

Happy Learning

THANK - YOU